

GigResource — Geographic matching developer response

19 August 2026 · UNDER REVIEW · Q5 / Q6 / Q7 / Q9 not locked · not built

Please stamp each question **APPROVE** · **MODIFY** · **REJECT** · **NEED MORE INFORMATION** before this is treated as a locked rule or an implementation. Nothing below invents a missing number; those items are marked NEED MORE INFORMATION.

PM stamps: Q5 _____ Q6 _____ Q7 _____ Q9 _____

Q5 — Professional Service Origin

Preference on the table (not locked): one dedicated Service Origin per Professional account. Travel radius is measured from it. It is not automatically the billing address.

Location fields that exist today

There is **one address block** on the Professional account. There is **no** billing street, **no** map coordinates, **no** travel radius, and **no** Service Origin field.

Field	Purpose today	Feeds search / matching?
Street (business address)	Professional business address. Used for address checks. PO Boxes are rejected. Paid address confirmation is built but switched off until launch.	No
City	Public profile. Homepage city counts. Browse city filter.	Yes — coarsely (name match, not miles)
State	Registered jurisdiction of this account. Locked once set (one account per state). Also decides whether the account may transact in the launch area.	Yes — today’s eligibility rule (same state as the event)
Country	With state, decides supported vs coming soon.	Indirect (transaction gate)
ZIP	Stored on the profile. At client sign-up, ZIP is mapped to a launch state using the first three digits. That mapping is state only — not a point on a map.	Indirect
Service-area status	Derived: operate here vs coming soon. Out-of-area accounts can register and browse; they cannot transact.	Transaction gate, not travel origin
Address-check status	Whether the street was checked, flagged as looking like a home, locked, etc.	No
Trade-licence state	Which launch jurisdiction issued the licence. Must match this account’s state.	No
Billing / card address	Membership billing is handled by the payment provider. GigResource does not store a separate billing street.	Does not exist here
Map coordinates, travel radius, Service Origin	Do not exist.	Do not exist

On the **request** (not the Professional): a free-text location (“city, venue, or address”) plus a two-letter event state. Same-state matching uses the event state. Miles cannot be calculated from free text.

Recommendation (pending stamp — not a schema lock). Create a dedicated Service Origin. Do not reuse the business street as the travel-radius origin, and do not treat it as billing (billing street is not in this product). Keep state as the same-state gate. Keep city and ZIP for display, directory counts, and ZIP fallback. V1 needs only one active origin: one account is already one state;

a second state is a second account; no live workflow needs two dispatch bases on the same account. Exact new field names are not proposed here.

NEED MORE INFORMATION (do not assume): default travel radius in miles and whether the Professional may edit it; whether the origin street is shown publicly or used only for distance math; confirmation that Service Origin may differ from the verified business address (recommended: yes).

Q6 — Distance calculation for matching

Preference on the table (not locked): straight-line / geodesic for marketplace search and matching. Driving distance, travel time, and travel fees can be a later feature.

Topic	Developer answer
Stack	The live product is Laravel on PHP, with MySQL in production and SQLite in tests. There is no PostGIS.
Method	Geodesic (great-circle) distance on stored coordinates. Production uses MySQL’s sphere-distance function (or the same Haversine formula in SQL). Tests use the same formula in PHP. This is straight-line, not road miles and not drive time.
When coordinates are created	Once, when the Professional origin or the event location is saved. Search and bidding never call a mapping service.
Indexing / speed	Same-state filter first (already indexed), then a bounding box on coordinates, then the precise circle check on the remaining rows. Launch is seven jurisdictions. With indexes this stays interactive at thousands of Professionals and acceptable into the tens / low hundreds of thousands in one state.
Mapping API for matching?	No.
API for saving an address?	Yes — a geocoder on save only. Primary: U.S. Census Geocoder (no per-request fee). Optional paid Google only if Census fails and a launch key already exists. Never on every browse or bid.
Costs	Search: \$0 per query. Census on save: \$0 (public; rate-limited on write). Google fallback: only if switched on, billed per save. ZIP centre table: \$0, public data for the seven launch states.
Trade-offs	Straight-line is shorter than driving. A Professional 20 road-miles away via a river or beltway may be 12 miles as the crow flies (included) or the reverse (excluded). That is the V1 trade-off requested. This is not travel time and not a travel fee. Same-state still wins: a circle that overlaps a neighbouring state does not match that state’s events.
Not recommended for V1 matching	Driving / directions lookups on each search (cost, delay, third-party dependency).

No extra information is required to approve the *method*. The default radius in miles is Q5-A, not a Q6 question.

Q7 — Geocoding failure and ZIP fallback

Proposed hierarchy (not locked): precise validated location → validated ZIP fallback → unresolvable. Matching must be checked on **both** sides: Professional origin and event location.

Fact today: GigResource can map a ZIP prefix to a state, not a ZIP to a point. ZIP fallback requires a ZIP centre table. Until that exists, a ZIP is not a coordinate. The event location is still free text.

Status	Meaning	Eligible to match?
Exact	Street or venue placed on the map.	Yes, if same state and within radius

ZIP	Validated ZIP centre stored; ZIP is not too broad.	Yes, with an accuracy warning
Unresolved	Neither exact nor an acceptable ZIP.	No. Fail closed. Never invent a point.

Situation	What the system does	What the person sees
Full address fails to geocode	Do not invent a point. If a valid 5-digit ZIP is in a launch state, has a centre, and is not too broad → save as ZIP. Otherwise unresolved. Matching stays off.	Client: “We could not place this location.” Not an empty professional list. Professional: origin matching is off, or “approximate (ZIP)” if ZIP was saved.
Only a valid ZIP	Map ZIP to launch state (already live). Look up the ZIP centre. Found and not too broad → ZIP status. Missing → unresolved.	“Location is approximate (ZIP). Nearby professionals are based on the centre of this ZIP, not a street.”
Unusually large ZIP	Proposed test (not locked): land area over 150 square miles, or the ZIP’s span is greater than half the travel radius (event side: greater than 15 miles if no radius applies) → unresolved. Ask for a city, venue, or street. Rural West Virginia and some Eastern Shore ZIPs will hit this on purpose.	Same as unresolved. Do not silently use the ZIP centre.
Neither coordinates nor acceptable ZIP	Unresolved. No coordinates. Not eligible. Request is not opened as an in-range gig. Do not fall back to “same city name.”	Client: “We could not place this location. Please enter a street, venue, or a more specific ZIP.” Professional: “Distance matching is off until this is fixed.”
Location placed, in-state, zero professionals in range	A real empty market.	“No professionals available in this area for this request.” Only this condition uses empty-market wording.

Silent eligibility is prevented by a hard gate: both sides must be exact or acceptable ZIP, both must have coordinates, same event state, and geodesic distance within radius. Unresolved or missing coordinates = not eligible. Tests must prove a failed geocode never appears in an in-range list.

NEED MORE INFORMATION (Q7-A): confirm or change the broad-ZIP cutoff (150 square miles / half of travel radius). Do not invent a different number without a stamp.

Q9 — Who counts toward the city directory

Approved rule: only Active and Eligible Professionals count. There is **no** Active / Suspended / Incomplete / Unapproved status on a Professional account. Influencers have pending / approved / rejected. Professionals do not. Eligibility is a set of flags.

Actual field	Counts on homepage today?	Count under Q9?
Professional role	Yes — required	Yes — required
Removed account	No	No
Locked out of sign-in (too many failed passwords)	Yes — gap	No (closest to Suspended)
Account deletion requested	Yes — gap	No

Coming soon (outside launch area)	Yes — gap	Only “supported” counts
Blank or non-launch state	Excluded	Launch state required
Blank city	Excluded from city pages	Required for a city page
Availability = not taking work	Yes — gap	Recommend exclude
Trade licence / insurance not verified	Yes — optional badges	NEED MORE INFORMATION — do not treat as Unapproved unless product locks that
No paid membership	Yes	NEED MORE INFORMATION (recommend No for V1)
Email not verified	Yes	NEED MORE INFORMATION (recommend No for directory)

Homepage today: Professional role + city + launch state. Threshold **2**, hardcoded. Q8 approved **5** as an admin setting — not built.

Recommended Q9 filter (pending stamp on Q9-A-D): Professional role; not removed; not locked out; not leaving; in a supported launch state; city present; not marked “not taking work.” Do not require badges or a paid plan unless stamped.

How this 13 August review maps to the live product

- **Approved and live:** Q1 / Q3 / Q4 same-state gate (Professional account state vs *event* state — two-letter state, not a numeric `state_id`). Q11 is true by default because radius matching does not exist yet.
- **Approved, not built:** Q2 radius (Option B: full circle, then drop other states). Q8 threshold 5 as an admin setting (live is 2, hardcoded). Q10 Coverage Map + 20-mile directory-only cap (live homepage is still base-city counts).
- **Still need a stamp:** Q5, Q6, Q7 (answered in this pack). Q9 mapping (this section).
- The review’s closing line that “every open item is resolved” covers Q2 / Q8 / Q10 / Q11. It does not close Q5–Q7.
- Do not build Q10 until Q5–Q7 are stamped — Coverage Map needs an origin, a radius, and a geocode failure state.